

Chapter 53

Keyframing Effects

The Edit and Cut pages also provide controls for keyframing effects that you add to your timeline.

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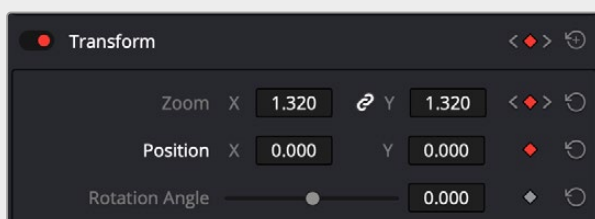
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EDIT

Keyframing Effects

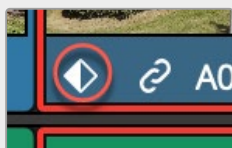
Most parameters in the Inspector can be keyframed, in order to create animated effects such as zooming in via the Zoom parameter, fading out via the Opacity parameter, or cropping from one side to reveal a clip underneath via the Cropping parameters. Additionally, if you import a project from an NLE that has keyframed sizing settings, those keyframes will be imported and exposed within DaVinci Resolve.

The basic controls for keyframing are within the Video Inspector. Any parameter that can be keyframed has a gray keyframe button to the right of its slider. If the playhead is on a keyframe, this button turns orange and small navigation arrows appear to its right and left, otherwise it stays gray.



Orange buttons in the Inspector show keyframe usage. Zoom shows the playhead parked on the current keyframe with additional keyframes set before and after this one as indicated by the gray navigation arrows. Position shows the playhead parked on the only keyframe set (orange diamond, no arrows), and Rotation Angle shows no keyframe set (gray diamond).

Once you've keyframed one or more parameters within a particular group in the Inspector, that clip displays a Keyframe icon at the far left of its name bar in the Timeline. Only keyframed clips have this icon.



The Keyframe track button (circled) in the Timeline, appearing on a keyframed clip

Basic Keyframing in the Video Inspector

Keyframing in the Cut and Edit pages works slightly differently than when using the Keyframe Editor in the Color page. Most simple keyframing tasks can be performed in the Inspector using three buttons that appear to the right of any parameter that's capable of being keyframed. It takes two keyframes at minimum to create an animated effect.



The three keyframe controls that appear in the Inspector, from left to right: Previous keyframe, Create/Delete keyframe, Next keyframe

Methods of keyframing parameters in the Inspector:

- **To add a keyframe:** Select a clip, open the Inspector, then move the Timeline playhead to the frame where you want to place a keyframe, and click the Keyframe button next to the parameter of the Inspector you want to animate. Once you've added at least one keyframe to a parameter, all other adjustments you make to parameters in the Inspector, or using the onscreen Transform/Crop controls in the Timeline Viewer add new keyframes automatically if the playhead is at another frame.
- **To move the playhead to the next or previous keyframe:** Click the small left- or right-hand arrow to either side of a parameter's keyframe control to jump the playhead to the next or previous keyframe. You can also press Right-Bracket (]) and Left-Bracket ([) to go from keyframe to keyframe.
- **To edit an existing keyframe of a parameter:** Move the playhead to be on top of the keyframe you want to edit, and then change that parameter, in the Inspector, or using the onscreen controls of the Timeline Viewer.

Methods of changing keyframe interpolation in the Inspector:

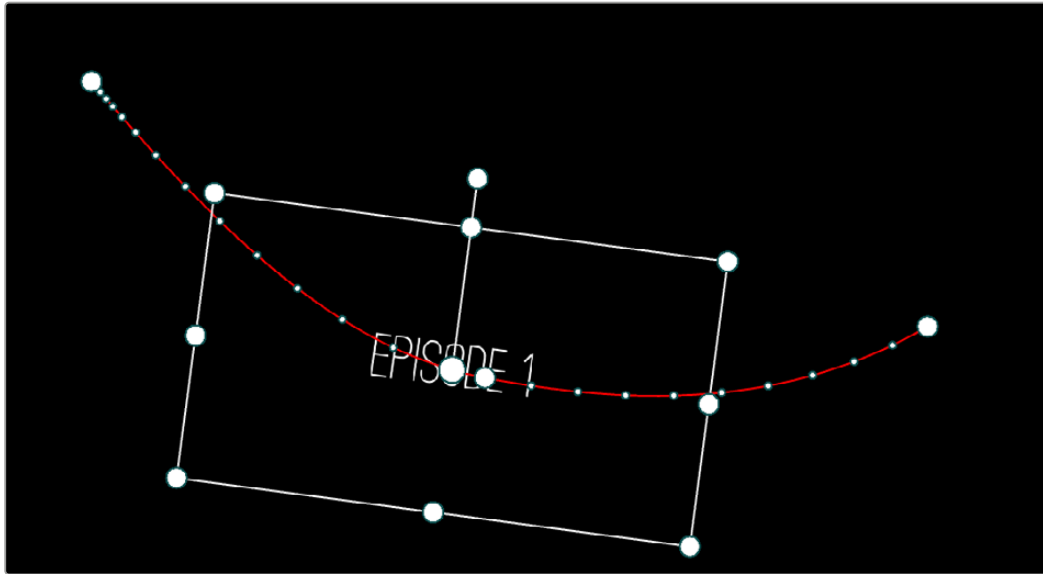
- **To change a keyframe to Ease In or Ease Out:** Eased keyframes create animated changes that begin slowly and accelerate to full speed, or slow down gradually to decelerate to a stop. This only works when you have two or more keyframes creating an animated effect. Move the playhead to a frame with a keyframe using the next/previous keyframe controls, then right-click the orange keyframe button and choose Ease In, Ease Out, or Ease In and Out, depending on which keyframe you're editing and the effect you want to create.
- **To change a keyframe to Linear:** Move the playhead to a frame with a keyframe using the next/previous keyframe controls, then right-click the orange keyframe button and choose Linear.

Methods of deleting keyframes and disabling keyframed effects:

- **To delete a single keyframe:** Open the Inspector, move the Timeline playhead to a frame with a keyframe, and click the orange Keyframe button in the Inspector to delete it.
- **To delete all keyframes for one parameter:** Click the reset button to the right of a parameter's keyframe control in the Inspector.
- **To delete all keyframes in a group of parameters in the Inspector:** Click the reset button to the right of a parameter group's title bar in the Inspector.
- **To disable or enable a group of parameters in the Inspector:** Click the toggle control at the left of a parameter group's title bar in the Inspector. Orange means that group is enabled. Gray is disabled.

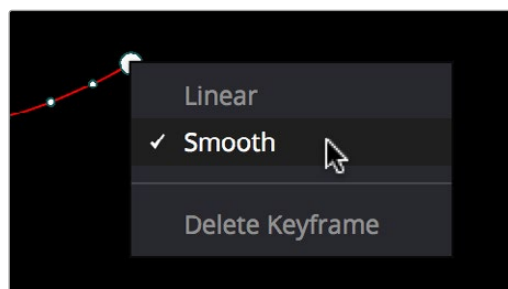
Keyframing Motion Paths in the Timeline Viewer

If you're keyframing a clip's transform controls to create motion, a motion path appears when you turn on the Transform controls using the drop-down menu at the lower left of the transport controls.



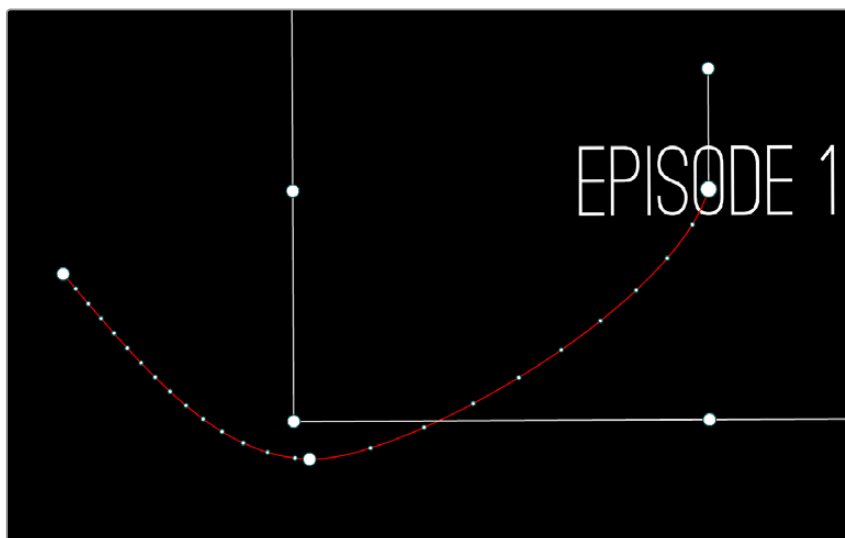
A visible motion path resulting from animated Position X and Y parameters

Each keyframed change to the Position X and Y parameters creates a control point on the surface of the motion path, which is linear by default, creating a sharp edge. However, you can right-click any control point and choose Smooth from the contextual menu to add Bezier handles to that control point, which let you change the sharp angle to an adjustable curve.



Changing the linear control point into a Bezier curve

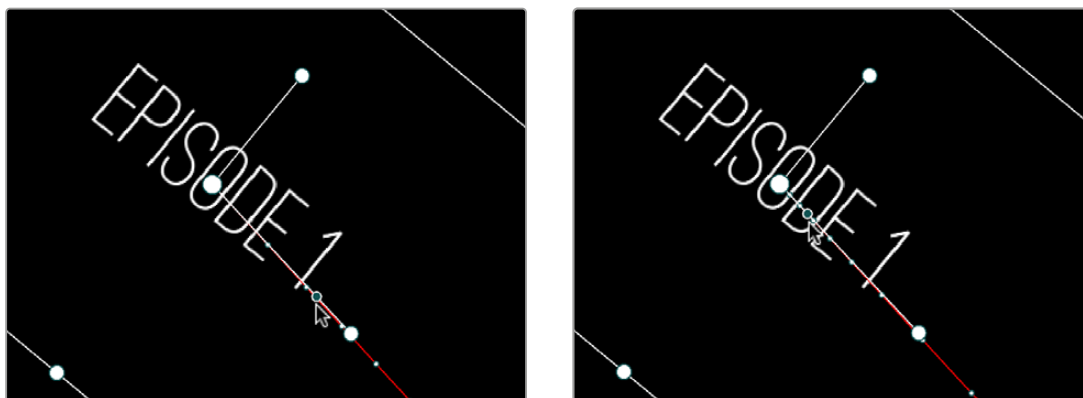
The control points making up any motion path can be dragged around at will to change the path the selected clip will travel. Dots on the surface of the motion path indicate the velocity of motion; dots that are closer together indicate slower motion, while dots that are farther apart indicate faster motion. Dragging a motion path control point farther away from another one will speed up the animation between both points, while dragging it closer will slow the animation down, as you're setting up the selected clip to travel a longer or shorter distance within the same keyframed time.



Dots on the motion path show that the left half has slow motion, while the right half has faster motion

You can also adjust the shape of any control point's curve by clicking to select that control point, which exposes its Bezier handles, and then dragging the handles to adjust its curve. Once handles have been exposed, there are a variety of methods you can use to adjust them and manipulate the motion path.

Finally, you can adjust the acceleration of motion by adjusting the Acceleration handle on the stem of any Bezier curve. Dragging an acceleration handle towards a control point creates an eased keyframe, where motion slows to a stop, or begins from a stop. Dragging an acceleration handle away from a control point creates more linear motion, where the object moves continuously through that control point.



An acceleration handle on the Bezier handle of a curve lets you create eased motion by dragging it in towards the control point being adjusted

Methods of adjusting the Bezier handles of motion paths:

- Drag any control point to reshape the motion path.
- Drag any Bezier handle to change the shape of the curve.
- Command-drag any Bezier handle to break the tangent between it and the opposite Bezier handle. When you release the Command key, the two Bezier handles become locked together again at whatever angle you created.

To eliminate a control point on a motion path, along with its keyframe:

- Right-click any control point and choose Delete Keyframe.

To switch a control point between sharp and curved angles:

- Right-click any control point and choose Linear (for a sharp angle) or Smooth (for a curve).

Keyframing in the Edit Timeline and Curve Editor

You can use keyframes directly in the the Timeline, or if you need to do more complicated keyframe editing than the relatively simple controls of the Inspector and Timeline allow, you can use the Keyframe tracks and Curve Editor found in the Edit Timeline. When one or more clip parameters are keyframed, two small buttons appear at the far right of a clip's name bar in the Timeline, a Curve button and a Keyframe button. These buttons let you access specialized keyframe editors that serve different purposes.

Keyframe Directly in the Timeline During Playback

Keyframes can be added directly on clips in the Timeline while playing back. You can add and delete keyframes to a clip using keyboard shortcuts, and DaVinci Resolve will try to intelligently predict the context of the clip to set the appropriate key frame.

This method allows you to quickly and precisely add retime and audio gain Keyframes during playback and then come back and manipulate the points you specified.

To Add a Keyframe to a Clip in the Timeline:

- Playback the clip on the Timeline and when the playhead is over the frame of the clip where you want to add the keyframe.
- Choose Mark > Add Keyframe (Command-[).)

To Delete a Keyframe in a Clip in the Timeline:

- Select the keyframe or keyframes you want to delete from the clip.
- Choose Mark > Delete Keyframe (Option-]).

While adding a keyframe on the timeline clip is a simple keyboard press, how does DaVinci Resolve know what specific parameter from all the options available in the inspector you wanted to keyframe? Essentially the keyframes are now context aware, meaning which keyframable attribute (Zoom, Position, Volume, etc.) is selected by DaVinci Resolve depends largely on the last item you manipulated, or falling back to the most commonly used.

The keyframe selection hierarchy in DaVinci Resolve runs through the keyframable options from top to bottom; if no keyframe makes sense in the context of the top entry, it automatically chooses the next one down the list, and so on.

The DaVinci Resolve Keyframe Selection Order:

- The Retime Controls (if already active)
- The active effect curve in the Keyframe Editor (if already open)
- The last Inspector control that was manipulated
- Audio Gain

For example, if you previously adjusted the Zoom parameter on a clip, then added a new keyframe to the clip by pressing Option-[, it would bypass the Retime Controls, the Effect Curves if those controls were closed in the inspector, and add a new Zoom keyframe, as that was the last tool manipulated. This also means that when you add a keyframe to a basic clip on a timeline with no modifications, it adds an Audio Gain Keyframe.

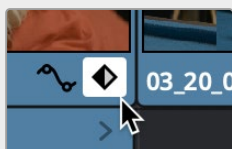
If you want to do more complex keyframing, or DaVinci Resolve is not picking up on the context correctly, you can use the Keyframe Editor.

The Keyframe Editor

The Keyframe Editor in the Timeline is the most powerful way of exposing all of a clip's keyframes and adjusting their timing and interpolation. It's only available when you've already keyframed one of a clip's Inspector properties.

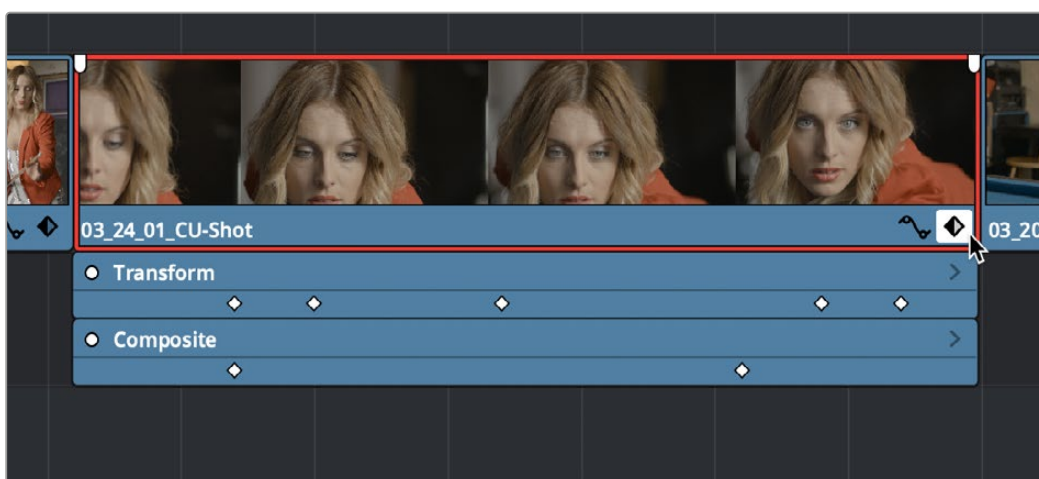
To open or close the Keyframe Editor:

- Click a clip's Keyframe button at the far right of a clip's name bar.
- Select a clip and choose Clip > Show Keyframe Editor (Command-Shift-C).



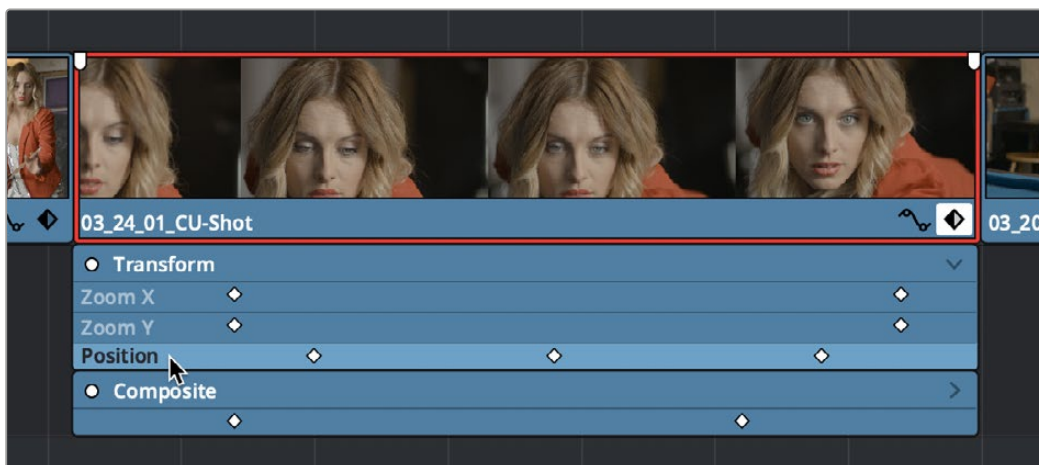
The Keyframe track button in the Timeline appearing on a keyframed clip

The Keyframe Editor exposes one keyframe track for each group of parameters that's keyframed. For example, the Composite parameters, Transform parameters, and Cropping parameters are all encapsulated by group tracks. For example, if you'd added keyframes to the Zoom and Position parameters, these keyframes all appear within a single keyframe track labeled Transform, while Opacity adjustments appear on a second keyframe track for Composite.



Group keyframe tracks open in the Timeline

However, each group keyframe track has a disclosure button that lets you show or hide each individual parameter that's keyframed within that group. For example, clicking the Transform keyframe track's disclosure button shows the Zoom and Position tracks, so you can adjust those individual keyframes.



Individual parameter keyframe tracks open in the Timeline

These keyframe tracks let you edit keyframes in context of the actual clip durations in the Timeline. Click the small Keyframe button at the bottom right of the clip's name bar to close the keyframe tracks when you're finished.

Methods of adding and selecting keyframes in the Keyframe Editor of the Edit page:

- **To add new keyframes to the Keyframe Editor:** Press Command-[or Option-click anywhere on a track of the Keyframe Editor to add a new keyframe, which defaults to whatever the current value is for that parameter at that frame. New keyframes create linear animated changes by default.
- **To duplicate one or more keyframes:** Make a selection of keyframes, then hold the Option key down and drag the selected keyframes to duplicate them and move the duplicates to a new position.
- **To select a single keyframe:** Click a single keyframe to select it.
- **To select multiple discontinuous keyframes:** Command-click all keyframes you want to select, whether they're next to one another or not.
- **To select multiple contiguous keyframes:** Click the first keyframe you want to select, and then shift-click the last keyframe you want to select, and all keyframes between will also be selected, or drag a bounding box within the keyframe track around multiple keyframes to select them all at once.

Methods of changing keyframe interpolation/easing/smoothing in the Keyframe Editor of the Edit page:

- **To change one or more Linear keyframe to Ease In or Ease Out:** Eased keyframes create animated changes that begin slowly and accelerate to full speed, or slow down gradually to decelerate to a stop. This only works when you have two or more keyframes creating an animated effect. Select one or more keyframes, then right-click one of the selected keyframes and choose Ease In, Ease Out, or Ease In and Out, depending on which keyframe you're editing and the effect you want to create.
- **To change one or more eased keyframes to Linear:** Select one or more keyframes, then right-click one of the selected keyframes and choose Linear.

Methods of moving and adjusting keyframes in the Keyframe Editor of the Edit page:

- **To move one or more keyframes:** Select one or more keyframes and drag left or right. While you drag keyframes, a tooltip appears showing you the offset in frames of your adjustment from the beginning of that clip's source media. If you're only dragging one keyframe, the tooltip also shows you the name of the parameter you're modifying.

- **To nudge selected keyframes one frame at a time:** Select one or more keyframes and press Command-Left Arrow or Command-Right Arrow to nudge them back and forth, for precision editing. The Curve Editor must also be open.

Methods of Cutting, Copying, Pasting, and Deleting keyframes:

- **To cut or copy, and paste one or more keyframes:** Make a selection of keyframes, and use the Cut (Command-X) or Copy (Command-C) key shortcuts. Then, move the playhead to where you want the first of the copied keyframes to start, and press Paste (Command-V). The Curve Editor must also be open.
- **To delete one or more control points from a curve:** Select the keyframe(s) you want to delete and press Backspace, or press Option-]. The Curve Editor must also be open.

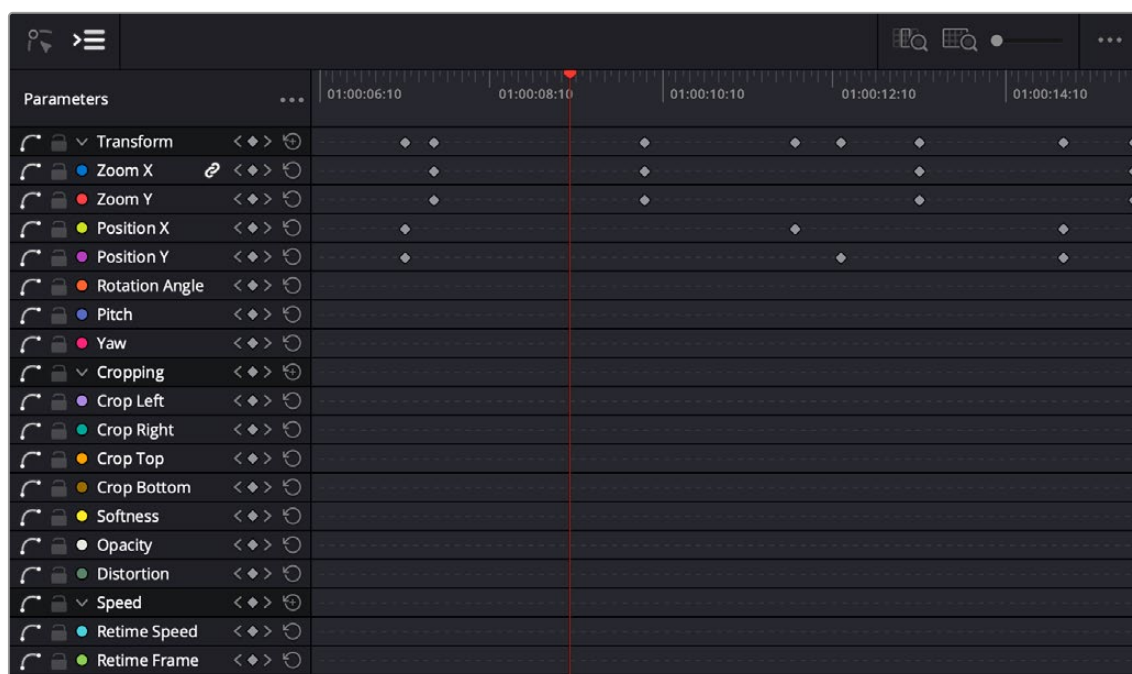
Keyframe and Curve Editor

The Keyframe and Curve Editors in DaVinci Resolve are designed to give you fine control for complex animations that are too detailed to use in the Inspector.

The Keyframe and Curve Editor is accessible from both the Cut and Edit pages as its own panel. Simply click on the Keyframes tab in the upper left to open it. By default, the Keyframe Editor is displayed, showing you all of the keyframable parameters in the Video Inspector and any existing keyframes that may have been added there. Clicking on the Curve icon in the upper left will take you to the Curve Editor.

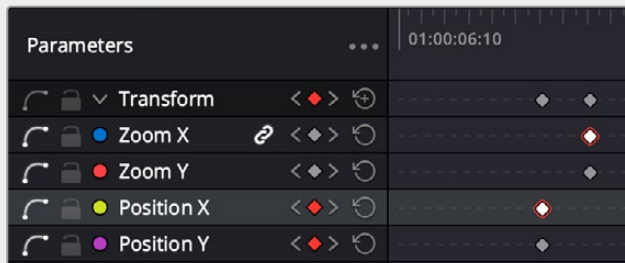
The Keyframe Editor

The Keyframe Editor in the Timeline is the most powerful way of exposing all of a clip's keyframes and adjusting their timing and interpolation.



The Keyframe Editor showing all the Video attributes found in the Inspector

The Keyframe Editor exposes one keyframe track for each parameter that's possible to be keyframed on the selected clip. Initially this looks like a big list of complicated items to keep track of (no pun intended), but it does provide a good overview of everything that's animatable for your clip. However, the Keyframe Editor has several tools to help remove unnecessary clutter and only focus on the exact parameters you're interested in animating. These keyframe tracks let you edit keyframes in context of the actual clip durations in the Timeline.



Keyframe Tracks: Transform is the track group (shown by the disclosure triangle), Zoom and Position XY are tracks within that group. Selected keyframes are shown white with an outline.

The tracks are also grouped by parameter category. For example, all the Transform, Cropping, and Speed parameters are encapsulated by group tracks. If you'd added keyframes to the Zoom and Position parameters, these keyframes also appear within a single keyframe track labeled Transform, while Crop Left and Crop Right adjustments appear on another group keyframe track labeled Cropping. Moving a keyframe on a group track will adjust the keyframes at the same position for all of its enclosed parameters at the same time.

Each group keyframe track has a disclosure arrow that lets you show or hide each individual parameter within that group. For example, clicking the Transform keyframe track's disclosure arrow shows the Zoom and Position tracks, so you can adjust those individual keyframes.

Organizing the Keyframe Editor

While keyframing itself is a fairly simple process, the sheer number of keyframable attributes can be overwhelming, so unless you're planning on animating something incredibly complicated, it's probably a better idea to focus the Keyframe Editor's interface on just what you need.

To show or hide tracks by parameter group

- Click on the small disclosure triangle next to the group track's name. For example, if you were only intending to keyframe parameters in the Transform group (Zoom, Position, etc.), you could close the disclosure triangles in the Cropping and Speed groups to hide the excess tracks.
- Click on the 3-dot option menu in the right of the parameters column, and choose Expand All Parameters to open every group track to see their enclosing parameters.
- Click on the 3-dot option menu in the right of the parameters column, and choose Collapse All Parameters to close all group tracks.

To show only tracks that already have keyframes

- Click on the 3-dot option menu in the right of the parameters column, and choose Display Parameters with Keyframes. This will show only tracks that have already at least one keyframe assigned to them, and their enclosing group tracks.

To show only Video related tracks

- Click on the 3-dot option menu in the right of the parameters column, and choose Display All Video Parameters. This will hide any keyframe tracks that control audio-related functions from the Audio Inspector.

To show only specific tracks

- Click on the 3-dot option menu in the right of the parameters column, and choose Display Selected Parameters. You can then choose checkboxes for Video and Audio. Clicking on these will open up another set of checkboxes that allow you to specify exactly which tracks you want to show.

To undock and resize the Keyframe Editor

- Click on the undock icon in the Keyframe Editor to open it as an independent window. You can then resize the window to expand the Keyframe Editor to any size that you need.

Using the Keyframe Editor

Once the Keyframe Editor is open and the parameters you wish to animate are exposed, you have many ways to add, delete, and manipulate keyframes along these tracks.



The Keyframe Track controls (L-R): Toggle Curve Visibility, Toggle Keyframe Lock, Track Color, Track Name, Keyframe, Reset Keyframes, Keyframe position.

Keyframe Track Controls:



Toggle Curve Visibility: Click on this icon to turn this track's curve visibility on or off in the Curve Editor.



Toggle Keyframe Lock: Click on this icon to turn the keyframe lock on or off to prevent accidental changes.



Track Color: A small colored dot shows the color of the track's curve in the Curve Editor.



Track Name: The name of the track parameter.



Keyframe: Clicking the diamond will add a new keyframe at the current position. If there is an active keyframe at the position this diamond will be orange, otherwise white. The left and right arrows will skip to the previous and next keyframes respectively.



Reset Keyframes: Removes all keyframes in the track.



Keyframe Position: Shows all set keyframes for the track and their position in the clip.

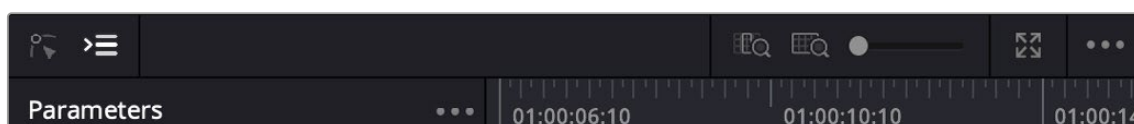
Methods of adding and selecting keyframes in the Keyframe Editor:

- **To add new keyframes to the Keyframe Editor:** Adjust the playhead to where you want the keyframe, then click on the gray diamond next to the parameter's name. The diamond will turn orange to let you know a keyframe has been added, and it will also appear on the parameter track at the playhead position. Subsequent keyframes can be added on this track by pressing Command [to add another keyframe.
- **To select a single keyframe:** Click a single keyframe to select it.
- **To select multiple discontinuous keyframes:** Command-click all keyframes you want to select, whether they're next to one another or not.
- **To select multiple contiguous keyframes:** Drag a bounding box within the keyframe track around multiple keyframes to select them all at once.
- **To move one or more keyframes:** Select one or more keyframes and drag left or right.

Methods of Cutting, Copying, Pasting, and Deleting keyframes:

- **To cut or copy, and paste one or more keyframes:** Make a selection of keyframes, and use the Cut (Command-X) or Copy (Command-C) key shortcuts. Then, move the playhead to where you want the first of the copied keyframes to start, and press Paste (Command-V).
- **To delete one or more keyframes:** Select the keyframe(s) you want to delete and press Delete.

IMPORTANT: Keyframes on the Timeline can exist past a clip's current extents. For example, if you set several keyframes on a clip, then trim its duration on the Timeline past one of the keyframes, that keyframe is still there and fully functional, just not visible. You can still navigate to these invisible keyframes by using the Previous "[" and Next "]" keyframe commands or using the keyframe controls in the Inspector.



The Keyframe Editor controls (L-R): Curves Editor, Keyframe Editor, Detail Zoom, Full Extend Zoom, Zoom Slider, Expand, Option Menu.



Curves Editor: Click this icon to open the Curves Editor.



Full Extend Zoom: Zooms the Keyframe Editor out to show the range of the whole clip.



Parameters: Click this icon to open the Keyframe Editor.



Zoom Slider: Lets you choose a custom zoom level for the Keyframe Editor



Detail Zoom: Zooms the Keyframe Editor in around the playhead position.



Expand: Opens the Keyframe and Curves Editor in its own resizable window.



Option Menu:

- **Display Time Ruler in Seconds:** Shows the time ruler in HH:MM:SS:FF timecode format based on the timeline.
- **Display Time Ruler in Frames:** Shows the time ruler as number of frames since the timeline start.
- **Snap Keyframes to Grid:** When enabled in the Curve Editor, this snaps keyframe control points to the playhead position.
- **Show Curve Tooltips:** When enabled in the Curve Editor, placing the pointer over a curve will show its track name and value at the given point.

Parameters

Parameters: Shows a list of keyframable parameters.

Option Menu:

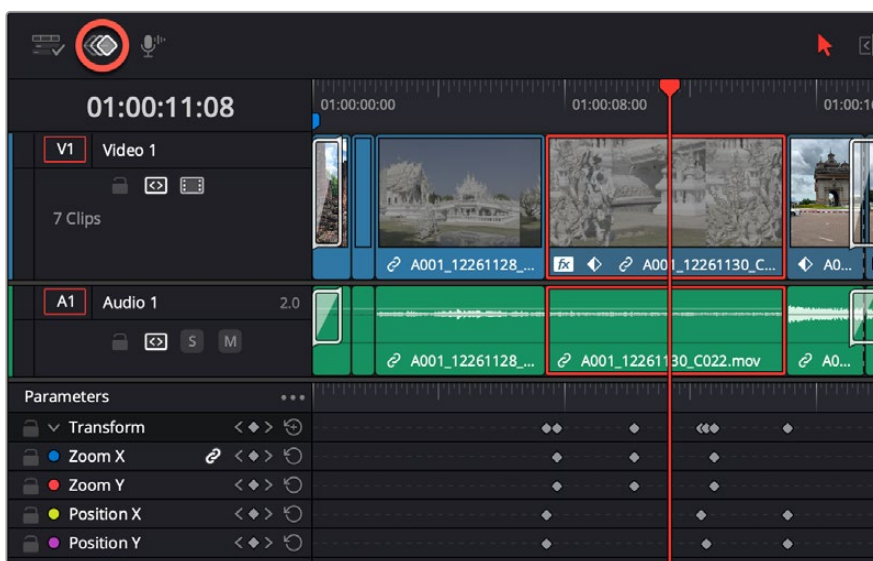
- **Display Parameters with Keyframes:**
Only shows tracks with existing keyframes and hides all others.
- **Display All Video Parameters:** Shows all keyframable tracks found in the Video Inspector, and hides all others.
- **Display Selected Parameters (Video & Audio):** Opens a checkbox to turn on or off individual track visibility. In addition, if you've added a keyframable Open FX or Resolve FX to the clip, you can select those parameters here.

Expand All Parameters: Opens every group track to see their enclosing parameters.

Collapse All Parameters: Closes every group track to hide their enclosing parameters.

Opening the Keyframe Tray in the Timeline

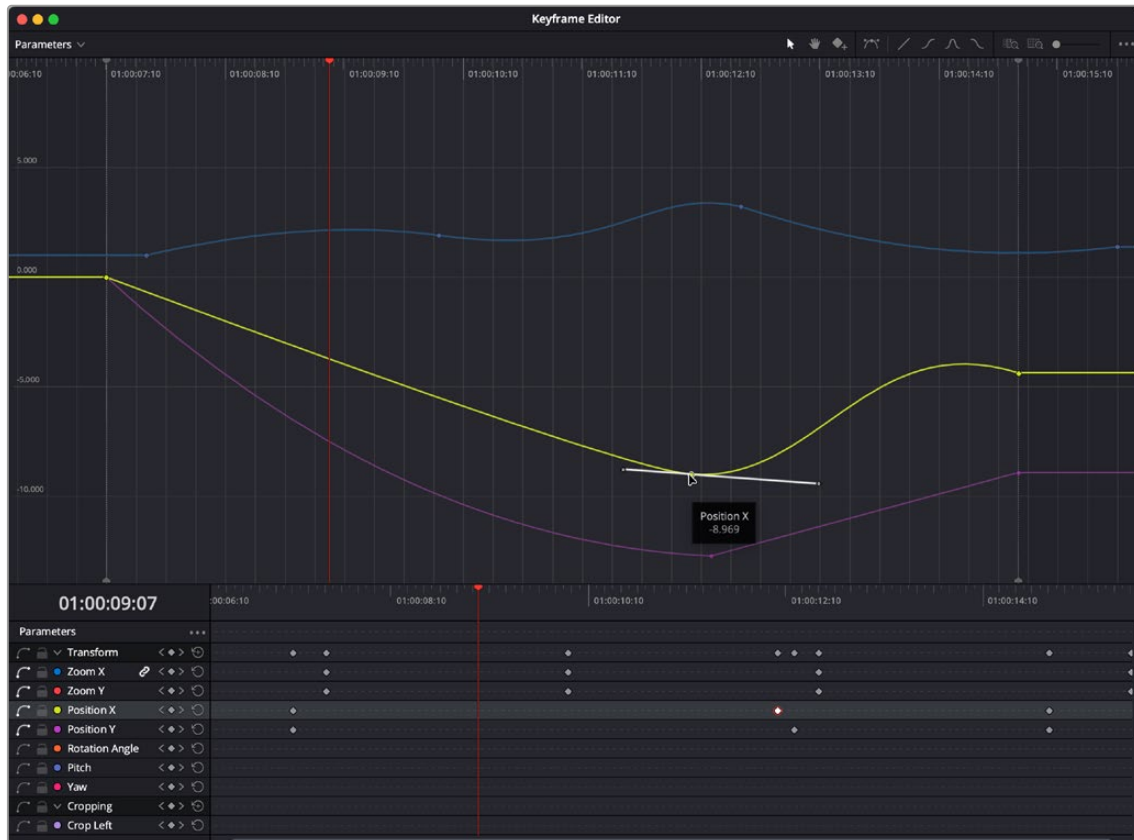
In addition to the dedicated Keyframe Editor panel, you can also expose a keyframe tray on the timeline by selecting the Show Keyframe Tray icon on the main timeline toolbar in both the Cut and Edit pages. This keyframe tray shows all the keyframes of the currently selected clip on the timeline and operates in the same way as the Keyframe Editor.



You can open up the Keyframe Editor directly in the timeline by clicking on the Show Keyframe Tray icon (circled red).

Opening the Keyframe and Curve Editor in Its Own Window

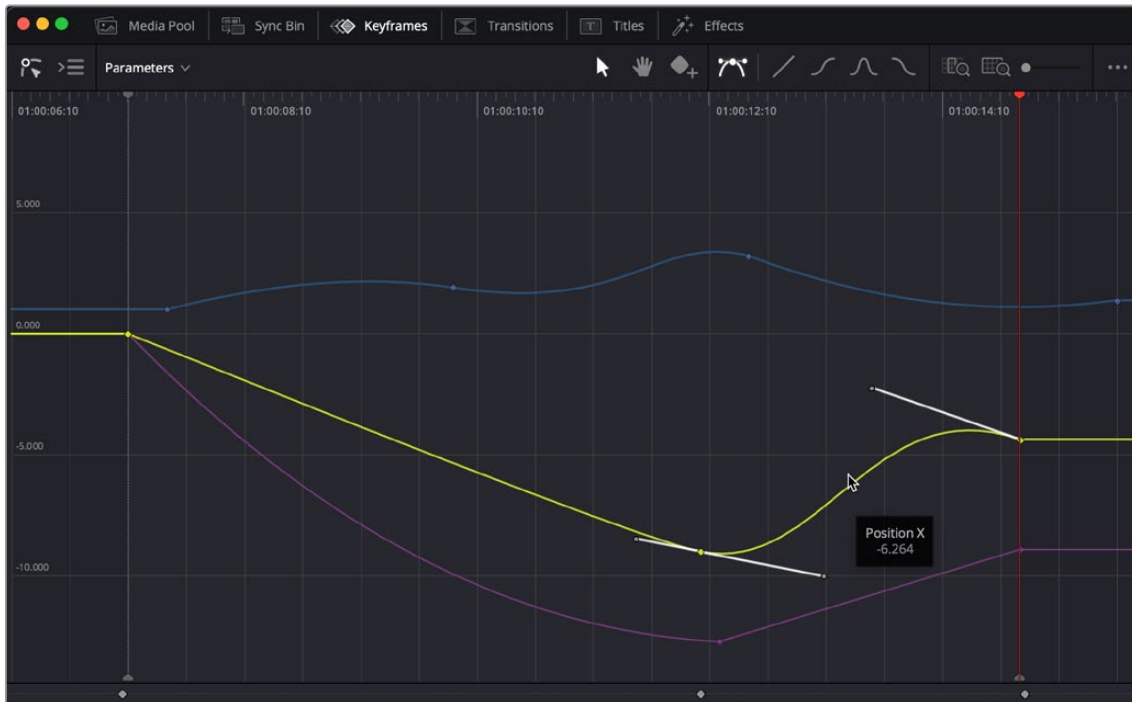
If you are doing extensive keyframing in the Edit page (not available in Cut), you can open up the Keyframe and Curve editor in its own resizable window to give you more space to work with. To do so, click on the Expand icon in the upper right of the Keyframe Editor.



The Keyframe and Curve Editor can be expanded to its own resizable window.

The Curve Editor

If you want to work with keyframes in even more detail, you can use the Curve Editor. When clicked, the the Curve Editor expands to accommodate a large graph in which you can freely adjust both the timing and value of selected keyframes, while also providing optional Bezier spline controls used to create smooth curves with which to adjust the acceleration of animated changes from one value to another.



The Curve Editor showing the timing and values of selected parameters.

To open the Curve Editor:

Click on the Curve icon in the upper left of the Keyframe Editor.

When opened, any track that has keyframes will appear as a colored line on the graph, with control points along the line at the keyframe position. You can hover the pointer over a line to display its track name (and its current value) in a pop up. The line color will also match the track color in the Keyframe Editor. You can open and close multiple parameters into the Curve Editor using the Parameters menu that lets you choose which tracks are exposed via checkboxes. At the bottom of the Curve Editor is a keyframe track identical to the Keyframe Editor that displays the keyframes for the selected curve.

While you can only work on one curve at a time, you can choose which is selected for editing by clicking any dimmed curve in the Curve Editor. Using the control points exposed by each curve, you can edit parameters, alter keyframe timing, and change each control point's interpolation to create custom easing effects affecting the acceleration of change from one keyframe to the next.

Methods of adding and selecting keyframes in the Curve Editor:

- **To change which curve you're editing:** Click the Parameters menu and choose which curves you want to expose to work on. If multiple curves are open in the Curve Editor, click any dimmed curve in the background to highlight it for editing.
- **To add new keyframes to a curve:** Option-click anywhere on a curve to add a new control point. You can also select the Add Keyframes icon at the top of the toolbar, which will let you add multiple keyframes to a curve, just by clicking on it.
- **To duplicate one or more keyframes:** Make a selection of keyframes, then press Command-C to copy, move the playhead to where you want to paste them and press Command-V. This can be a good way to quickly loop a repetitive animated effect you've created.

- **To select a single keyframe:** Click a single control point to select it. You can perform the same operation on the keyframe track at the very bottom of the Curve Editor.
- **To select multiple discontinuous keyframes:** Command-click all control points you want to select, whether they're next to one another or not. You can perform the same operation on the keyframe track at the very bottom of the Curve Editor.
- **To select multiple contiguous keyframes:** Click the first keyframe you want to select, and then shift-click the last keyframe you want to select, and all keyframes between will also be selected, or drag a bounding box within the Curve Editor around multiple keyframes to select them all at once. You can perform the same operation on the keyframe track at the very bottom of the Curve Editor.
- **To select all keyframes:** If the Keyframe Editor is open and it has focus (by clicking anywhere within it), then pressing Command-A will select all keyframes within that Keyframe Editor.

Methods of adjusting keyframes in the Curve Editor:

- **To drag one or more keyframes freely on a curve:** Select one or more control points and drag left or right to retime them, and up or down to change their value. A popup will display the track name and the updated value as you move it.
- **To drag one or more keyframes on a curve in only one direction:** Select one or more control points, then hold the Shift key while dragging either vertically or horizontally to constrain keyframe adjustment within that single direction.

Methods of changing keyframe interpolation/easing/smoothing:

- **To change one or more Linear keyframe to Ease In or Ease Out:** Eased keyframes create animated changes that begin slowly and accelerate to full speed, or slow down gradually to decelerate to a stop. This only works when you have two or more keyframes creating an animated effect. Select one or more keyframes, then choose Ease In, Ease Out, or Ease In and Out from the toolbar, depending on which keyframe you're editing and the effect you want to create.
- **To change one or more eased keyframes to Linear:** Select one or more keyframes, then choose Linear from the toolbar.
- **To change the interpolation of multiple keyframes:** Select multiple keyframes by Command-clicking or dragging a bounding box, and then click one of the four Bezier interpolation buttons in the Curve Editor title bar to simultaneously change the interpolation of all of them.
- **To adjust a Bezier handle:** Drag the Bezier handle in any direction to alter the curve.

Methods of Cutting, Copying, Pasting, and Deleting keyframes:

- **To cut or copy, and paste one or more keyframes:** Make a selection of keyframes and use the Cut (Command-X) or Copy (Command-C) key shortcuts. Then, move the playhead to where you want the first of the copied keyframes to start, and press Paste (Command-V).
- **To delete one or more keyframes from a curve:** Select the keyframe(s) you want to delete and press Delete.

Curve Editor Controls



The Curve Editor controls (L-R): Curves Editor, Keyframe Editor, Parameters, Selection, Hand, Add Keyframe, Linear, Ease in, Ease out, Ease in and Out, Detail Zoom, Full Extend Zoom, Zoom Slider, Expand, Option Menu.



Curves Editor: Click this icon to open the Curves Editor.



Keyframe Editor: Click this icon to open the Keyframe Editor.



Parameters: Opens a checkbox to turn on or off individual curve visibility. In addition, if you've added a keyframable Open FX or Resolve FX to the clip, you can select those parameters here.



Pointer Mode: Returns to the standard mode where clicking on things with the pointer selects them.



Hand Mode: Lets you click and drag to navigate through the Curve Editor when zoomed in. Useful in highly detailed and complicated animations.



Add Keyframes: Turns the pointer into an Add Keyframe tool that will add a keyframe on any curve in the Curve Editor, simply by clicking it.



Show Handle: Toggles the Bezier handle controls on or off.



Linear: Sets the curve to a direct linear progression from keyframe to keyframe.



Ease In: Sets the curve to gradually intensify the curve from the starting keyframe.



Ease In and Out: Sets the curve to gradually intensify the curve from the starting keyframe and gradually intensify the curve towards the ending keyframe.



Ease Out: Sets the curve to gradually intensify the curve towards the ending keyframe.



Detail Zoom: Zooms the Keyframe Editor in around the playhead position.



Full Extend Zoom: Zooms the Keyframe Editor out to show the range of the whole clip.



Zoom Slider: Lets you choose a custom zoom level for the Keyframe Editor.



Expand: Opens the Keyframe and Curves Editor in its own resizable window.

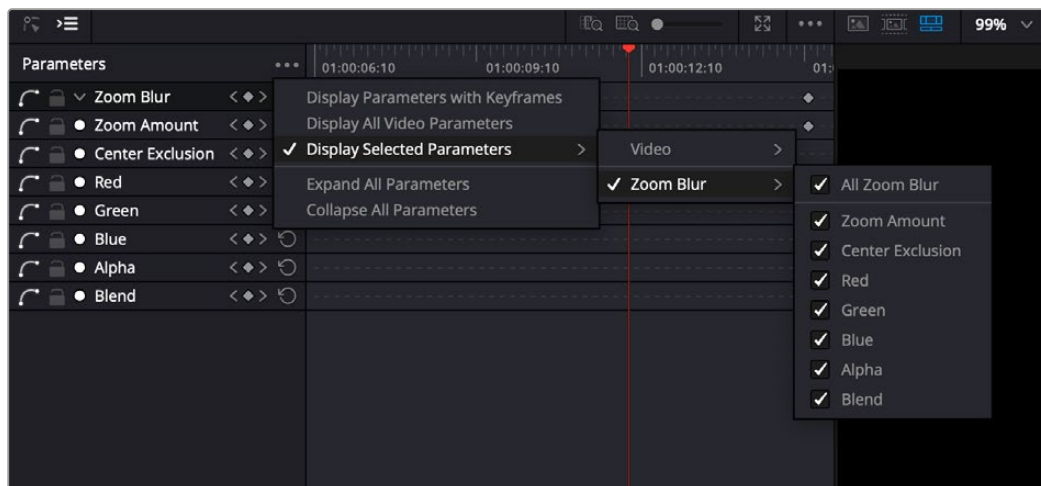


Option Menu:

- **Display Time Ruler in Seconds:** Shows the time ruler in HH:MM:SS:FF timecode format based on the timeline.
- **Display Time Ruler in Frames:** Shows the time ruler as number of frames since the timeline start.
- **Snap Keyframes to Grid:** When enabled in the Curve Editor, this snaps keyframe control points to the playhead position.
- **Show Curve Tooltips:** When enabled in the Curve Editor, placing the pointer over a curve will show its track name and value at the given point.

Keyframable Open FX and Resolve FX

Keyframes added to Resolve FX parameters appear in both the Keyframe and Curve Editor. Using the Display Selected Parameters menu in the 3-dot option menu, you can expose individual keyframe tracks and curves for each keyframed parameter of an effect applied to a clip, for smoothing, retiming, or editing.



The Open FX and Resolve FX can be adjusted in the Keyframe and Curves Editor by selecting it in Display Selected Parameters.